AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

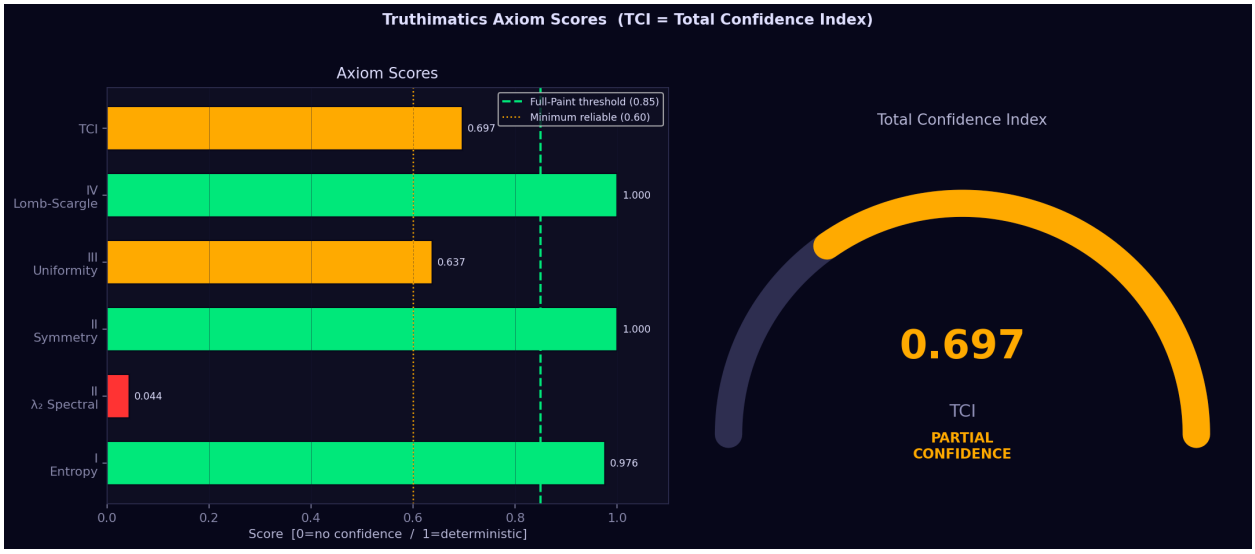
Specimen ID	AXIOM_CA7FA1FE22
Taxa	Euoplocephalus
Clade	Ornithischian
PBDB Records	18
Mass (regression)	5778.0 kg (Campione & Evans 2012)
Mass (volumetric)	1511.9 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	1.591 m³
Validity Score	1.0000 / 1.0000
TCI	0.6966 (partial)
Axiom I Entropy	0.9759
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.6375
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.234, 0.466, 0.0]
Generated	2026-04-30 23:59 UTC

Ratio Violations

- ilium/femur=1.232 outside [0.45,0.90]
- ischium/femur=0.884 outside [0.35,0.75]

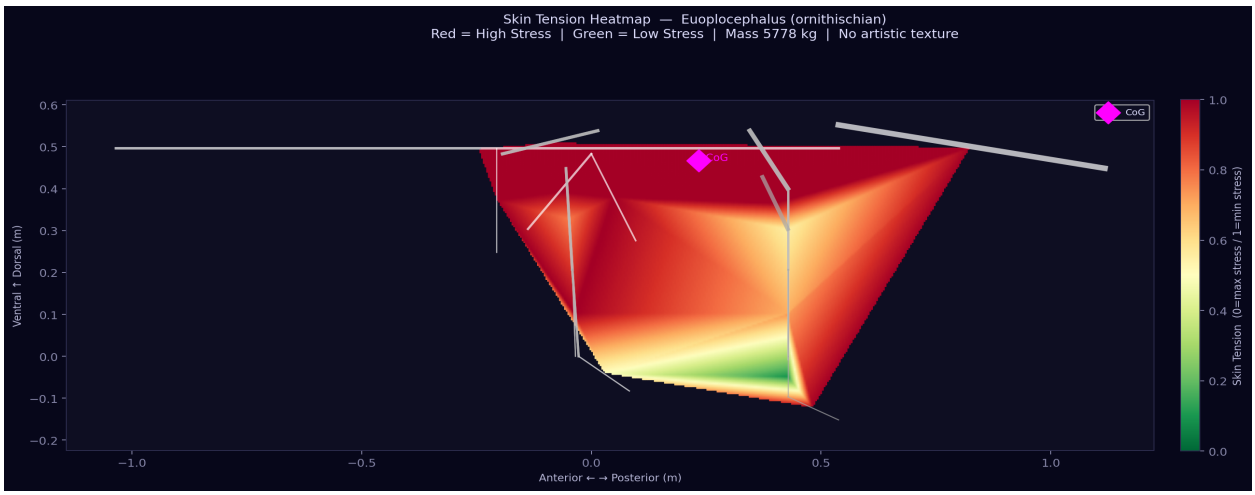
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



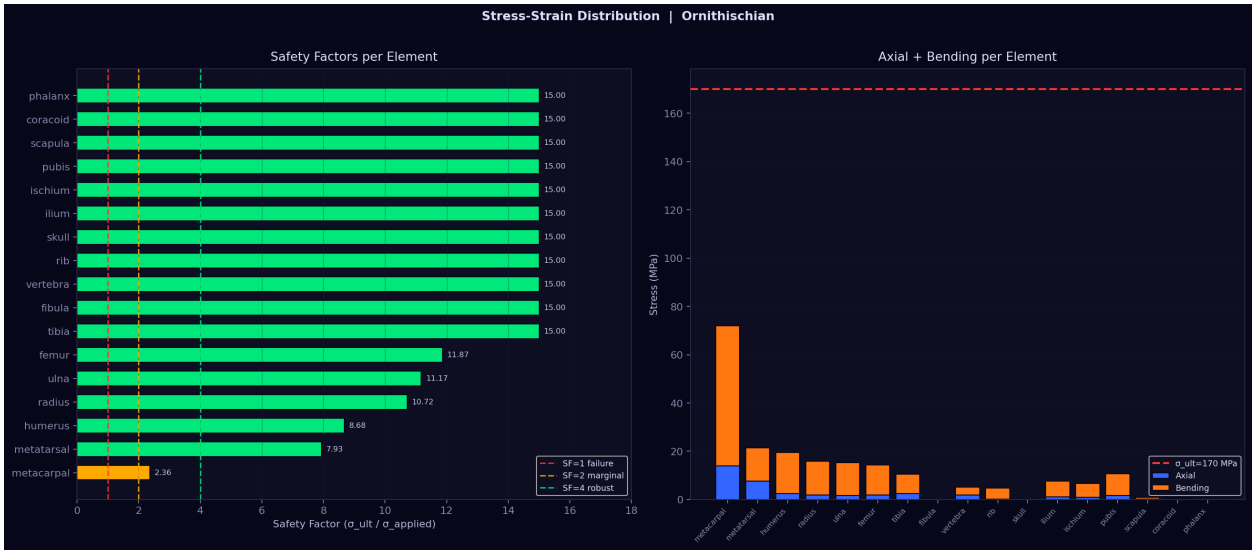
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

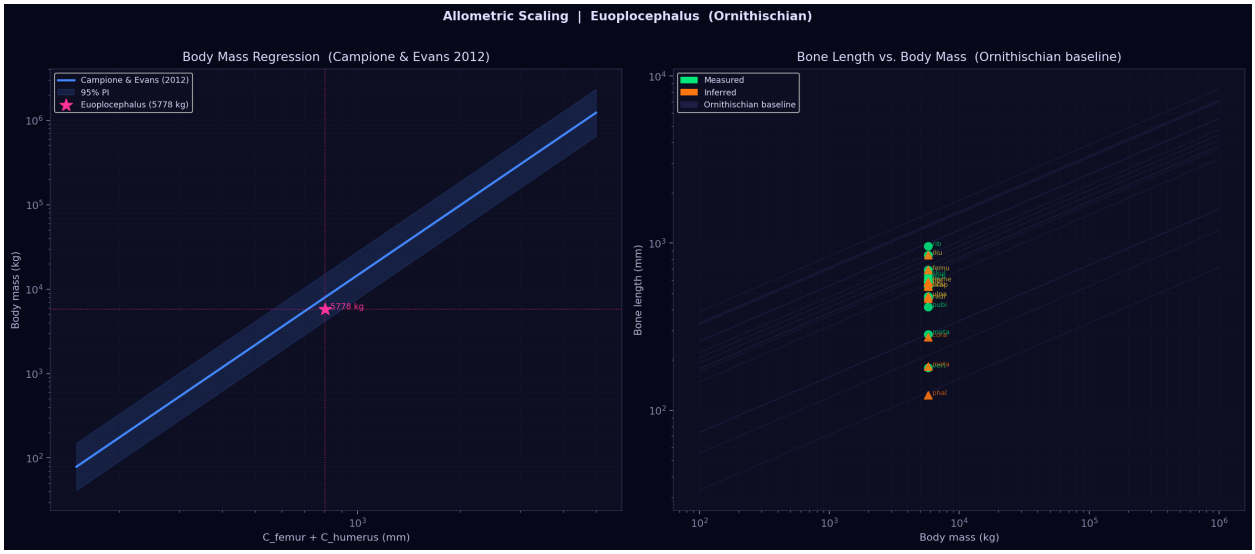


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.284	0.500	0.784	15.000	OK
femur	1.858	12.468	14.325	11.867	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	2.541	17.052	19.593	8.677	OK
ilium	1.196	6.346	7.542	15.000	OK
ischium	0.987	5.773	6.760	15.000	OK
metacarpal	13.978	57.923	71.900	2.364	WARNING
metatarsal	7.651	13.773	21.424	7.935	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	1.714	8.985	10.699	15.000	OK
radius	1.841	14.017	15.858	10.720	OK
rib	0.175	4.673	4.848	15.000	OK
scapula	0.238	0.768	1.007	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	2.541	8.046	10.587	15.000	OK
ulna	1.755	13.460	15.215	11.173	OK
vertebra	1.858	3.252	5.110	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	1.555 m	Femur + tibia + metatarsal
Estimated Speed	11.6 km/h (3.22 m/s)	Gait: Walk | Fr=0.678
Stride Length	3.175 m	Alexander (1976) / Thulborn (1990)
Bite Force	6989.2 N (712.5 kgf)	Ornithischian regression (Norman & Weishampel 1991)
Skull Length	640.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.6283	Circumference / length (dimensionless)
Limb Strength Index	4e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	6.3 m	Allometric from femur length
Est. Neck Length	0.69 m	Allometric from femur length
Est. Tail Length	1.76 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	274.0	208.8	4785.06	5973.2	Constraint	0.40	15.00
femur	690.0	138.0	5263.41	6570.3	Measured	0.88	11.87
femur_right	690.0	138.0	5263.41	6570.3	Symmetry	0.84	11.87
fibula	555.0	50.0	392.31	411.5	Measured	0.88	15.00
fibula_right	555.0	50.0	392.31	411.5	Symmetry	0.84	15.00
humerus	590.0	118.0	3290.61	4107.7	Measured	0.88	8.68
humerus_right	590.0	118.0	3290.61	4107.7	Symmetry	0.84	8.68
ilium	850.0	172.0	10072.47	12573.5	Measured	0.88	15.00
ilium_right	850.0	172.0	10072.47	12573.5	Symmetry	0.84	15.00
ischium	610.0	112.0	3064.97	3826.0	Measured	0.88	15.00
metacarpal	182.9	48.8	64.89	53.4	Constraint	0.65	2.36
metatarsal	285.0	68.0	527.86	658.9	Measured	0.88	7.93
phalanx	123.3	41.8	25.94	20.1	Constraint	0.40	15.00
pubis	415.0	85.0	1201.01	1499.2	Measured	0.88	15.00
radius	465.0	82.0	1252.39	1563.4	Measured	0.88	10.72
radius_right	465.0	82.0	1252.39	1563.4	Symmetry	0.84	10.72
rib	960.0	58.0	1293.56	1614.8	Measured	0.88	15.00
scapula	548.0	228.0	11410.65	14243.9	Measured	0.88	15.00
scapula_right	548.0	228.0	11410.65	14243.9	Symmetry	0.84	15.00
skull	640.0	488.0	61049.16	76207.7	Measured	0.88	15.00
tibia	580.0	118.0	3234.83	4038.0	Measured	0.88	15.00
tibia_right	580.0	118.0	3234.83	4038.0	Symmetry	0.84	15.00
ulna	480.0	84.0	1356.63	1693.5	Measured	0.88	11.17
ulna_right	480.0	84.0	1356.63	1693.5	Symmetry	0.84	11.17
vertebra	180.0	138.0	1373.06	1714.0	Measured	0.88	15.00